

Z-score & Robust Statistics

วัดว่า spread ห่างจากค่ากลางแค่ไหน — Standard vs Robust

KEY IDEA

Z-score converts the spread residual ϵ into units of standard deviations from the mean, producing a dimensionless signal that drives entry and exit decisions.

$$\text{Standard } z = (\epsilon - \theta) / \sigma_{\text{stat}}$$

$$\text{Robust } z = (\epsilon - \text{median}) / (\text{MAD} \times 1.4826)$$

The robust version protects against single outliers distorting the signal — critical when exchange glitches or thin-liquidity spikes contaminate the feed.

10.1 Standard Z-Score

Given a rolling window of **W** observations of the spread residual ϵ , the standard z-score is:

$$z_t = (\epsilon_t - \theta_W) / \sigma_W \text{ where: } \theta_W = (1/W) \sum \epsilon_{\{t-i\}} \text{ for } i = 0 \dots W-1 \text{ (rolling mean)}$$

$$\sigma_W = \sqrt{(1/W) \sum (\epsilon_{\{t-i\}} - \theta_W)^2} \text{ (rolling std dev)}$$

A common parameterisation uses **W = 24 hours** of data (e.g. 288 five-minute bars). Entry and exit thresholds map z-score values to actions:

Z-score range	Signal zone	Action
$z > 2.0$	Strong long signal	Enter long ϵ (buy cheap leg, sell expensive leg)
$1.5 < z \leq 2.0$	Watch zone	Monitor; wait for confirmation
$0.5 < z \leq 1.5$	Neutral-high	Hold existing position
$-0.5 \leq z \leq 0.5$	Neutral / exit	Close position ($z < 0.5$)
$-1.5 < z < -0.5$	Neutral-low	Hold existing short position
$-2.0 < z \leq -1.5$	Watch zone	Monitor short signal
$z \leq -2.0$	Strong short signal	Enter short ϵ (sell cheap leg, buy expensive leg)
$ z > 3.5$	Stop-loss zone	Emergency exit — outlier or regime break

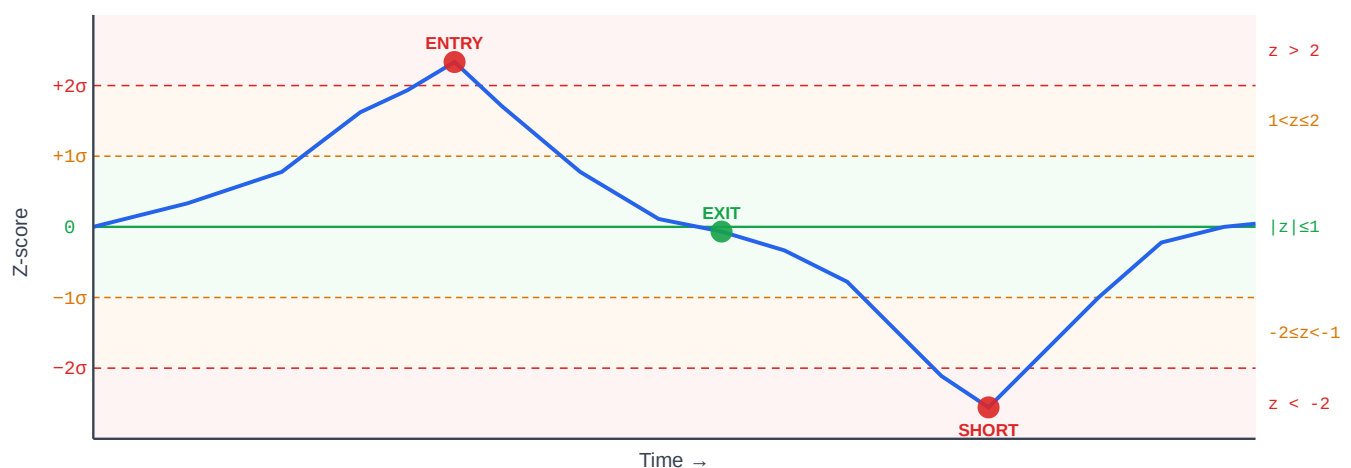


Figure 10.1 — Rolling z-score with colored signal bands. Blue curve = ϵ expressed in σ units. Red dot = entry signal ($z > 2$). Green dot = exit ($|z| < 0.5$).

10.2 The Outlier Problem

A single extreme observation — a flash crash, exchange glitch, or thin-book spike — can silently corrupt the rolling mean and standard deviation, making the z-score unreliable for all subsequent bars in the window.

Numerical Example

Suppose $W = 10$ bars with ϵ values (in %) before the outlier:

```
 $\epsilon = [0.10, 0.12, 0.09, 0.11, 0.10, 0.13, 0.08, 0.11, 0.10, 0.12]$   $\theta_{\text{clean}} = 0.106\%$   
 $\sigma_{\text{clean}} = 0.015\%$  z for next point  $\epsilon=0.22\%$ :  $z = (0.22 - 0.106)/0.015 = 7.6 \leftarrow$  strong signal
```

Now inject one outlier: replace bar 5 with $\epsilon = 0.80\%$ (5σ spike):

```
 $\epsilon = [0.10, 0.12, 0.09, 0.11, 0.80, 0.13, 0.08, 0.11, 0.10, 0.12]$   $\theta_{\text{dirty}} = 0.176\%$   
 $\sigma_{\text{dirty}} = 0.212\% \leftarrow$  mean shifted  $+0.07\%$ , std blown up  $14\times$  z for same  $\epsilon=0.22\%$ :  $z = (0.22 - 0.176)/0.212 = 0.21 \leftarrow$  noise, signal gone!
```

Warning: Outlier Contamination Effect

The outlier moved the mean by $+0.07\%$ ($+0.47\sigma$ of the clean distribution) and inflated σ by roughly $14\times$. Every subsequent observation now has a z-score compressed toward zero — real signals are suppressed, and the bands become meaningless until the outlier leaves the window.

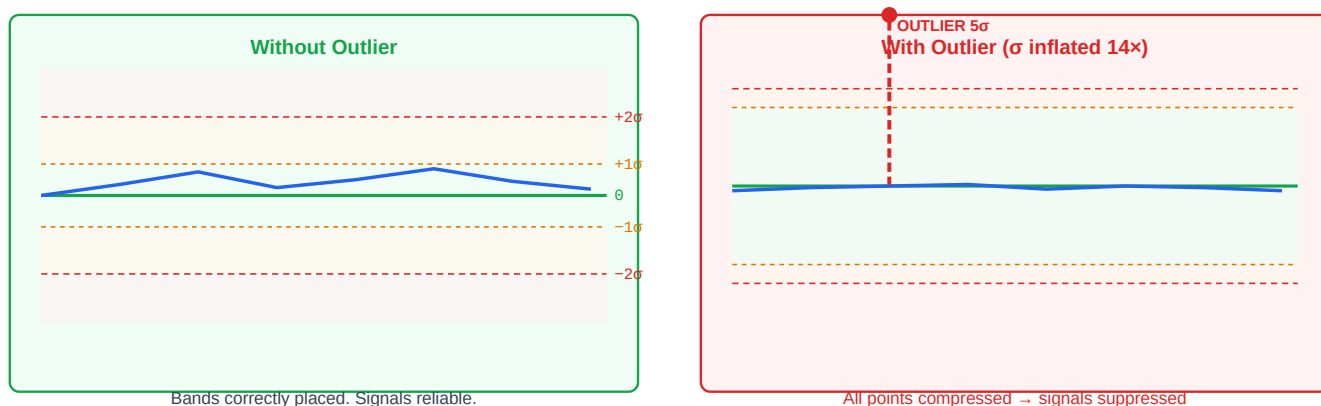


Figure 10.2 — Left: healthy z-score bands ($\sigma \approx 0.015\%$). Right: one 5σ outlier inflates σ and compresses all bands, hiding real signals.

10.3 Robust Z-Score via MAD

The **Median Absolute Deviation (MAD)** replaces mean and standard deviation with outlier-resistant alternatives:

```
Step 1: median_ε = median(ε1, ε2, ..., εW) Step 2: MAD = median( |εt - median_ε| )  
Step 3: σrobust = MAD × 1.4826 Step 4: robust_z = (εt - median_ε) / σrobust
```

Why 1.4826?

For a perfectly Normal distribution, $\text{MAD} \times 1.4826 = \sigma$. The constant $1.4826 = 1 / \Phi^{-1}(0.75) \approx 1 / 0.6745$ makes MAD a consistent estimator of σ under normality, while remaining robust under contamination. A single outlier cannot materially change the median or MAD — it would need to corrupt more than 50% of the data.

Revisiting the same example with the 5σ outlier injected:

```
ε = [0.10, 0.12, 0.09, 0.11, 0.80, 0.13, 0.08, 0.11, 0.10, 0.12] median_ε = 0.11%  
deviations from median = [0.01, 0.01, 0.02, 0.00, 0.69, 0.02, 0.03, 0.00, 0.01, 0.01]  
MAD = median(deviations) = 0.01% σrobust = 0.01 × 1.4826 = 0.01483%  
robust_z for ε=0.22%: (0.22 - 0.11) / 0.01483 = 7.42 ← signal preserved! (vs  
standard z = 0.21 after outlier contamination)
```

Result

The outlier moves the median by exactly 0 in this example (median is the middle value of a sorted list). Robust z correctly identifies ε=0.22% as a strong signal (z=7.4), while standard z falsely showed no signal (z=0.21).

10.4 Rolling Window Z-Score

Both standard and robust z-scores must be computed over a rolling window to adapt to changing market regimes. The window length **W** is a key hyperparameter:

Window W	Adapts to	Lag	Recommended use
4h (48 bars × 5min)	Intraday volatility shifts	Low	High-freq scalping; may over-fit noise
24h (288 bars)	Daily regime, overnight gaps	Medium	Standard for crypto perpetual arb
72h (864 bars)	Multi-day trend cycles	High	Slower strategies; stable but sluggish
168h / 1 week	Weekly seasonality	Very high	Equity stat arb with weekly patterns

Lag Warning During Trend

If the spread drifts monotonically for several hours (e.g. during a BTC rally), the rolling mean θ_W chases the price upward with a lag equal to $W/2$. This artificially suppresses z-score readings — the strategy may fail to trigger exit on a position that is actually losing value. Consider adding a trend filter or using exponentially-weighted moving averages (EWMA) with $\lambda = 0.94$.

10.5 Comparison: Standard vs Robust Z-Score

Property	Standard Z	Robust Z (MAD)
Formula	$(\varepsilon - \text{mean}) / \text{std}$	$(\varepsilon - \text{median}) / (\text{MAD} \times 1.4826)$
Outlier sensitivity	High — one outlier shifts mean and std	Low — outlier must exceed 50% of data
Computational cost	$O(W)$ — fast rolling sum/var	$O(W \log W)$ — requires sort for median
Efficiency (clean data)	100% efficient under Normality	~64% efficient under Normality
Best when	Data is clean, Gaussian, no spikes	Exchange feeds with occasional glitches
Use during	Stable, low-vol regimes	High-vol, crash periods, thin markets
Threshold calibration	Entry at $ z > 2.0$	Entry at $ z > 2.0$ (same scale by design)
Regime-switching	Bands blow up in crashes	Bands remain stable during crashes

Recommendation

Use **Robust Z as default** for crypto stat arb on Bybit/Lighter. Fall back to standard z only when your data pipeline has confirmed outlier filtering (e.g. price validated against two independent feeds).

Hybrid: use robust z for entry, standard z for exit (since exit thresholds are softer and less sensitive to contamination).

Pseudo-Code

```
function compute_zscore(eps, theta, sigma): # Standard z-score (single-point, pre-
computed  $\theta$  and  $\sigma$ ) if sigma == 0: return 0.0 return (eps - theta) / sigma
function rolling_zscore(eps_series, W): # Compute standard rolling z-score for full series
z_series = [] for t in range(W, len(eps_series)): window = eps_series[t-W : t] theta
= mean(window) sigma = std(window) z_series.append(compute_zscore(eps_series[t],
theta, sigma)) return z_series
function robust_zscore(eps_window): # Compute robust z
for the last point in eps_window med = median(eps_window) devs = [abs(x - med) for x
in eps_window] mad = median(devs) sigma_r = mad * 1.4826 if sigma_r == 0: return 0.0
return (eps_window[-1] - med) / sigma_r
function choose_signal(eps_window, W): # Use
both; flag discrepancy std_z = rolling_zscore(eps_window, W)[-1] rob_z =
robust_zscore(eps_window) discrepancy = abs(std_z - rob_z) if discrepancy > 1.5:
log("OUTLIER SUSPECTED - using robust z") return rob_z, "robust" return std_z,
"standard"
```

Running Example — Bybit BTC-PERP ↔ Lighter BTC-PERP (W = 24h)

Scenario: BTC crashes 12% over 2 hours at 03:00 UTC. Bybit order book thins out. One 5-minute bar records $\epsilon = 0.45\%$ (a feed glitch as a single large market order clears thin asks).

Metric	Standard Z	Robust Z	Outcome
Window mean / median	$\theta = 0.126\%$	median = 0.10%	Robust unaffected by spike
σ estimate	$\sigma = 0.060\%$ (inflated 4×)	$\sigma_r = 0.015\%$	Standard σ blown up
Z-score for next bar $\epsilon = 0.32\%$	$z = (0.32 - 0.126) / 0.060 = 3.2$	$z = (0.32 - 0.10) / 0.015 = 14.7$	—
Z-score for real signal $\epsilon = 0.22\%$	$z = 1.6$ (below entry threshold 2.0)	$z = 8.0$ (strong signal)	Standard misses entry
After outlier leaves window (24h later)	z recalibrates correctly	z unchanged (already correct)	Robust reacts immediately

Decision: During the BTC crash window, use **robust z for entry decisions**. The standard z produced a false alarm at $\epsilon=0.32\%$ ($z=3.2$ suggesting strong signal, but driven by inflated σ from the

spike) and missed the real signal at $\varepsilon=0.22\%$. The robust z correctly ranked signals by true deviation from median spread.

Exercises

Exercise 10.1 — Compute Robust Z

Given the following window of 7 ε values (all in %):

$\varepsilon = [0.10, 0.12, 0.08, 0.11, 0.80, 0.09, 0.11]$

Compute: (a) median, (b) MAD, (c) σ_{robust} , (d) robust_z for the *last point* $\varepsilon = 0.11\%$.

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Exercise 10.2 — Discrepancy Between Standard Z and Robust Z

You observe: **$\text{robust_z} = 2.3$** , **$\text{standard_z} = 4.1$** . Your entry threshold is $|z| > 2.0$.

(a) What does the discrepancy of 1.8 σ -units suggest? (b) Should you enter the trade? (c) What additional check would you perform?

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